

SINGLE-FAMILY LOAN PERFORMANCE · FANNIE MAE

Predicting mortgage delinquency, *six months early.*

A loan-level model that flags Fannie Mae mortgages likely to become **90+ days delinquent within a six-month window**, trained on a monthly panel of 12.5 million loan records and tuned to surface a short, workable watchlist.

12.5M

LOAN-MONTH ROWS

172

MODEL FEATURES

0.949

ROC-AUC · TEST

0.71

RECALL AT SHIP

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FINAL MODEL

Oversampled LightGBM

Report Contents

Thirteen sections trace the project end to end, from a raw stack of Fannie Mae acquisition and performance files to a calibrated, time-honest delinquency model and the operating point it ships at.

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Executive Summary

This project builds a model that flags Fannie Mae mortgages likely to become **90 or more days delinquent within the next six months**. It is trained on a monthly panel of **12,524,939 loan-month records** drawn from a 10% borrower sample of loans originated in the first half of 2021, with each loan followed month by month as it performs.

The final model, an **oversampled LightGBM classifier**, reaches a ROC-AUC of **0.949** on a held-out, time-separated test set of 2.47 million loan-months it never saw in training.

The central difficulty is rarity. Only about **three in every thousand loan-months** go on to serious delinquency, so raw accuracy is meaningless. A model that predicts "never delinquent" for every loan scores 99.7% accurate and is useless. Judged instead on the precision-recall curve, gradient-boosted trees pulled clearly ahead of linear and neural baselines.

At its shipped operating point the model **catches 71% of future-delinquent loans while flagging just 1.4% of the book**, turning millions of loan-months into a queue a servicing team can actually work.

"By the time a loan is 90 days past due, most of the good options are already gone. The useful question is the earlier one."

0.949

ROC-AUC ON HELD-OUT TEST

71%

OF FUTURE DEFAULTS CAUGHT

1.4%

OF THE BOOK FLAGGED

Problem Definition & Audience

WHY IT MATTERS

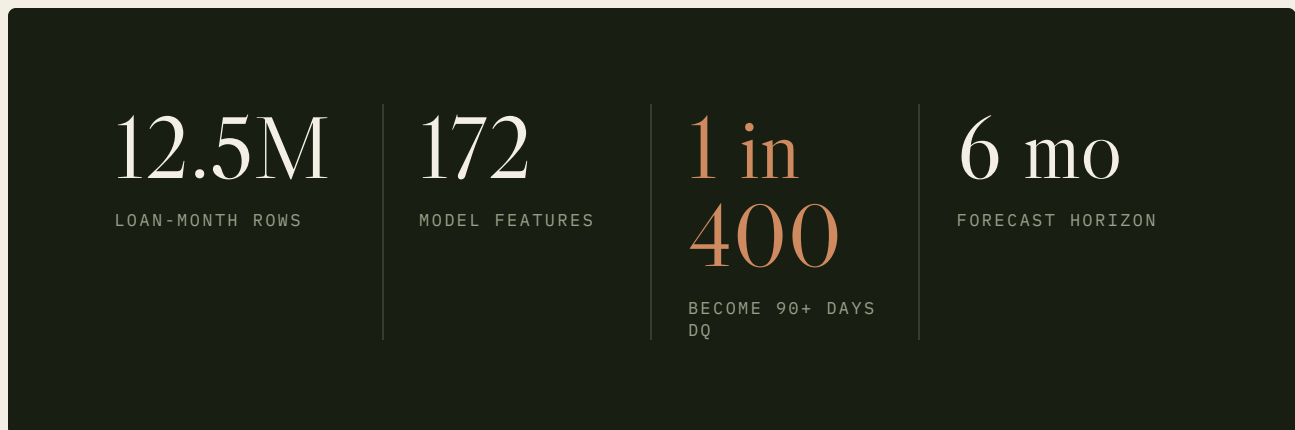
A mortgage that quietly slides toward delinquency is costly for everyone: the servicer, the investor, and the household. By the time a loan is 90 days past due, most of the good options are already gone.

THE QUESTION

Given everything we know about a loan this month, can we predict whether it becomes 90+ days delinquent within the next six months, early enough to actually act on?

WHO IT'S FOR

Servicing and credit-risk teams: the people deciding which borrowers to call first, where to target loss-mitigation outreach, and how to size reserves against the book.



The audience cares about three things: **catching as many future defaults as possible**, keeping the resulting watchlist small enough to work, and trusting that the model was scored honestly, on loans from a period it was never trained on.

Data Source & Structure

SOURCE

Fannie Mae's public **Single-Family Loan Performance** dataset, the standard benchmark for U.S. mortgage credit research. I worked from a 10% borrower sample of loans originated in **2021 Q1-Q2**, then followed each loan month by month as it performed.

WHAT'S IN A ROW

One row is **one loan in one reporting month**: its current balance and delinquency status, joined to fixed origination facts such as credit scores, DTI, LTV, interest rate, loan purpose, occupancy, property type, and geography.

SCALE

Followed across the performance window, the sample expands to a monthly panel of **12,524,939 labeled loan-month rows**.

SCHEMA · SELECTED FIELDS

loan_id	str
reporting_month	date
current_actual_upb	float
loan_dq_status	str
borrower_credit_score	int
dti	float
original_ltv / cltv	float
original_interest_rate	float
loan_purpose / occupancy	str
property_state / type	str
seller / servicer_name	str

Target Variable Definition

future_90plus_dq_6m

For every loan-month, I look **six months ahead**. If the loan reaches 90+ days delinquent at any point in that window, the row is labeled **1**; otherwise **0**.

Rows without a full six-month forward window were dropped, along with a small unlabeled tail. What remains is a heavily lopsided pair of classes, and that imbalance is the central challenge of the whole project.

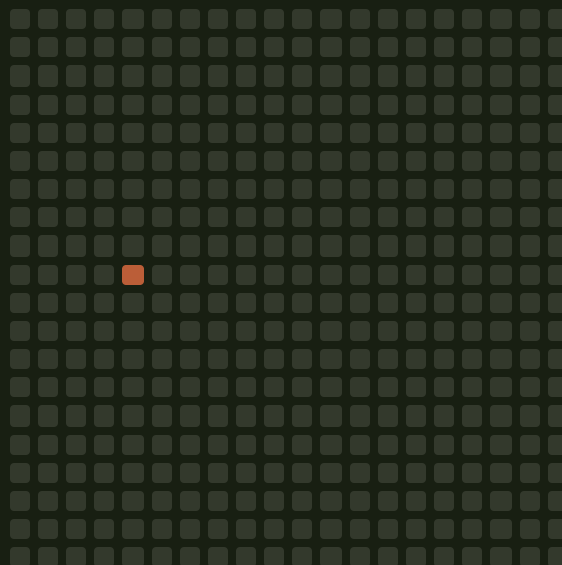
99.75%

NO FUTURE 90+ DQ

0.25%

FUTURE 90+ DQ (POSITIVE)

EACH SQUARE = 1 LOAN-MONTH · 400 SHOWN



■ The one that becomes seriously delinquent.

A NOTE ON RATES. Every rate in this report is at the loan-month grain. Across the full sample the positive rate is about 0.25%, roughly one row in 400, rising to 0.30% in the later, more-seasoned test window.

Building the Loan-Month Panel

RAW
Acquisition + Performance files



12.5M
LABELED LOAN-MONTH ROWS

01

JOIN ACQUISITION + PERFORMANCE

One record per loan-month, balance and status aligned to each loan's fixed origination terms.

02

STANDARDIZE & DOWNCAST

Memory-conscious dtypes, dates parsed once, fields renamed for clarity across 4.6 GB of records.

03

ENGINEER DELINQUENCY HISTORY

Rolling 12-month counts of 30+ and 60+ day past-due spells per loan, plus prior status and change flags.

04

BUILD THE FORWARD LABEL

Look six months ahead to set `future_90plus_dq_6m` for each row.

05

DROP THE UNLABELED TAIL

Remove rows lacking a full six-month forward window, so every label is real and complete.

Credit Score: The Primary Risk Driver

The 90+ day delinquency rate falls more than **thirty-fold** from the lowest origination credit-score band to the highest. It is the single strongest signal in the data, and reassuringly, it bends in exactly the direction underwriting intuition expects.

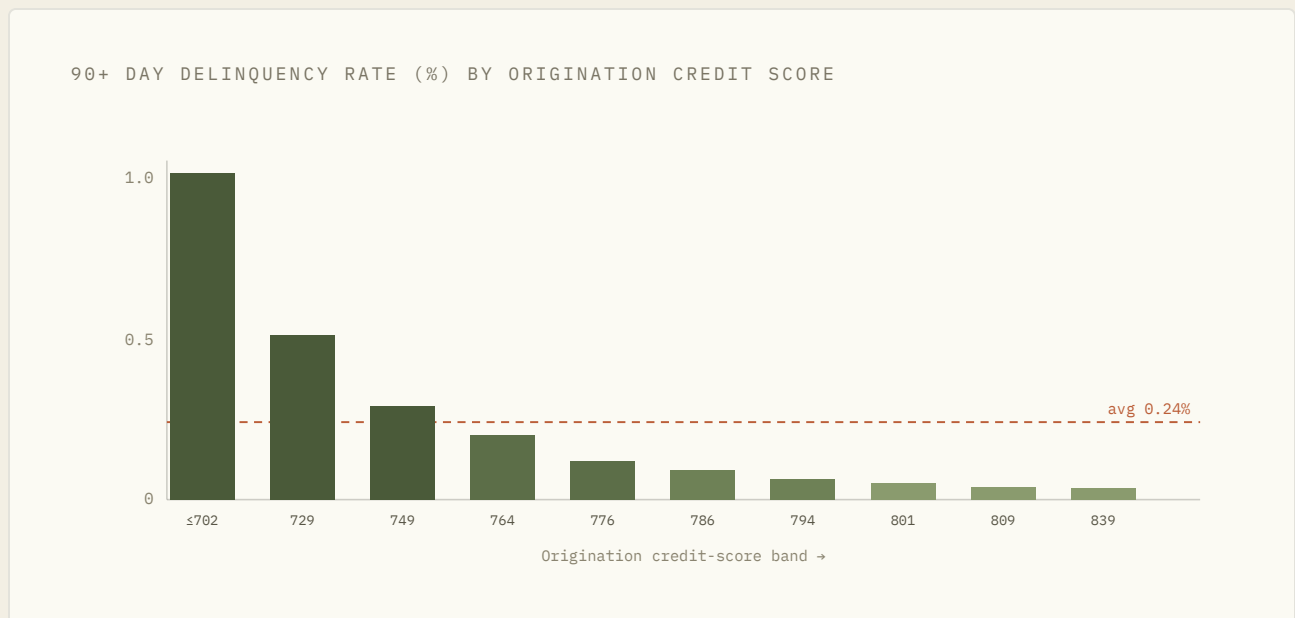
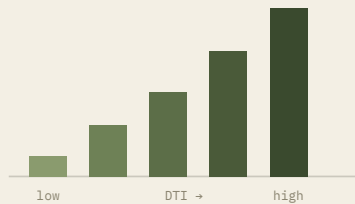


FIG 01. Delinquency rate by origination credit-score decile, at the loan-month grain. Bands left of the dashed sample-average line (0.24%) carry above-average risk; the gradient is monotone across the whole range, spanning more than thirty-fold from the lowest band (1.01%) to the highest (0.03%).

Leverage, Channel & Delinquency History

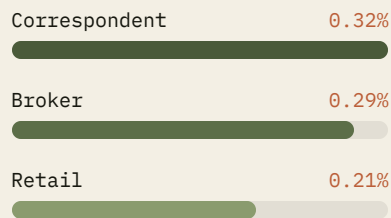
LEVERAGE RAISES RISK

Higher debt-to-income climbs steadily with delinquency, the mirror image of the credit-score curve.



CHANNEL MATTERS

Retail-originated loans run noticeably safer than broker and correspondent channels.



RECENT LATENESS SHOUTS

A loan already past due this month is the loudest near-term signal of where it heads next.

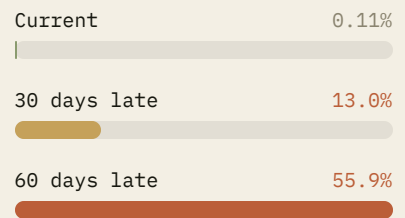


FIG 02-04. Loan-month delinquency rates by debt-to-income band, origination channel, and current delinquency status. Recent lateness dwarfs every static origination feature: the six-month probability of serious delinquency leaps from 0.1% for loans that are current to about 56% for loans already 60 days late.

Feature Construction & Preprocessing

ORIGINATION TERMS

Borrower & co-borrower credit scores, DTI, original LTV / CLTV, interest rate, loan purpose, occupancy.

BALANCE & PAYDOWN

Original and current UPB, UPB ratio, paydown %, total principal paid to date.

DELINQUENCY HISTORY

Rolling 30+ and 60+ day past-due counts over 12 months, prior status, status-change flags.

TIME CONTEXT

Loan age, months remaining to maturity, reporting month, quarter, and year.

172

MODEL FEATURES

The pipeline. One-hot encoding for categoricals → standard scaling for numerics → a 172-column feature matrix. Trees don't need the scaling, but the pipeline applies it uniformly so every model is compared on identical inputs.

Time-Ordered Train / Test Split

This is a forecasting problem, so the test set has to live in the **future** relative to training. A random split would let the model peek at later months while scoring earlier ones, inflating every metric. Instead the data is split **chronologically**: the earliest 80% of the timeline to train, the latest 20% held out and touched only once, at the very end.



WHY TIME-ORDERED

The held-out test block is strictly later than everything the model trains on, the same position the model would face in production, predicting months that haven't happened yet.

TOUCHED ONCE

Model selection and threshold tuning happen inside the training window only. The test set is scored exactly once, after every decision is locked.

Class Imbalance & Metric Selection

A model that simply predicts "never delinquent" for every loan scores **99.7% accurate** and is useless. So I oversampled the rare positive class during training, and judged models on the **precision-recall curve** rather than accuracy.

ROC-AUC barely separates the candidates: true negatives are so abundant that everything looks good. PR-AUC, measured against the tiny base rate, is the honest yardstick for a rare event.

~0.3%

POSITIVE BASE
RATE

PR-

AUC

PRIMARY METRIC

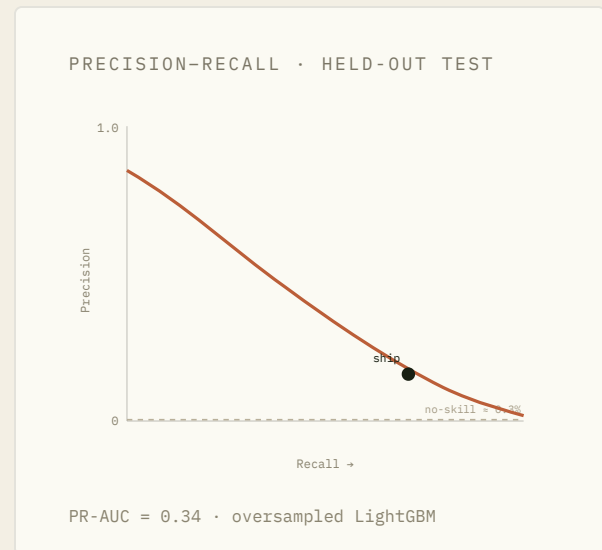


FIG 05. Precision-recall curve on the held-out test set. The oversampled LightGBM sits far above the no-skill floor that a 0.3% base rate implies; the marked point is the shipped operating threshold.

Candidate Model Comparison

MODEL	ROC-AUC	PR-AUC	RECALL	PRECISION
Oversampled LightGBM FINAL	0.949	0.340	0.75	0.121
Tuned XGBoost	0.948	0.337	0.82	0.044
Neural Network	0.941	0.329	0.26	0.534
Oversampled Logistic Reg.	0.942	0.224	0.72	0.132
Logistic Regression	0.943	0.213	0.82	0.034
Dummy (predict majority)	0.500	0.003	0.00	0.00

ROC-AUC clusters tightly between 0.90 and 0.95, so it can't tell these models apart. **PR-AUC can:** oversampled gradient-boosted trees pull clearly ahead.

The neural network buys precision by catching only a quarter of defaults; the linear models do the reverse. **Oversampled LightGBM strikes the best balance**, and trains fastest on data this size.

Metrics shown at the default 0.5 cutoff; the shipped operating point is tuned on the next page.

Operating-Threshold Selection

The model outputs a probability; the cutoff turns it into a flag. Because the goal is to catch as many future-delinquent loans as possible, I pushed the threshold up to **0.70**, accepting more false alarms to hold recall near 0.71.

Even at that aggressive setting, the model flags **under 1.4% of the book** for review, small enough for a servicing team to actually work.

0.70

CHOSEN THRESHOLD

0.71

RECALL HELD

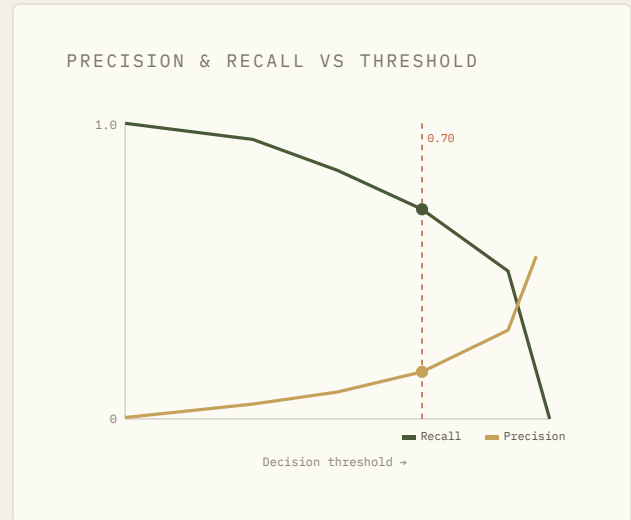


FIG 06. Precision and recall as the decision threshold sweeps from 0 to 1. The recall-first goal pushes the operating point to the right, trading precision for coverage, and the watchlist gets a little noisier, but very few true defaults slip through.

Held-Out Test Performance

0.949

ROC-AUC · HELD-OUT

0.71

RECALL

0.34

PR-AUC

1.36%

FLAG RATE

CONFUSION MATRIX · 2.47M TEST ROWS

TRUE POSITIVES

5,272

caught

FALSE POSITIVES

28,255

false alarms

FALSE NEGATIVES

2,113

missed

TRUE NEGATIVES

2.43M

correct passes

On 2.47 million loan-months it never saw in training, the model **catches 71% of the loans** that go on to serious delinquency while flagging just 1.4% of the book.

About **one flagged loan in six** is a true future default, more than fifty times the 0.3% base rate, and exactly the kind of lift that makes an outreach queue worth working.

Principal Model Drivers

The most-used features echo the EDA: creditworthiness at origination, leverage, and loan size. One caveat: because this is *split-count* importance, the binary recent-delinquency flags rank lower here than their predictive power warrants. No single exotic feature carries the model; it is familiar credit fundamentals, weighed together at scale.

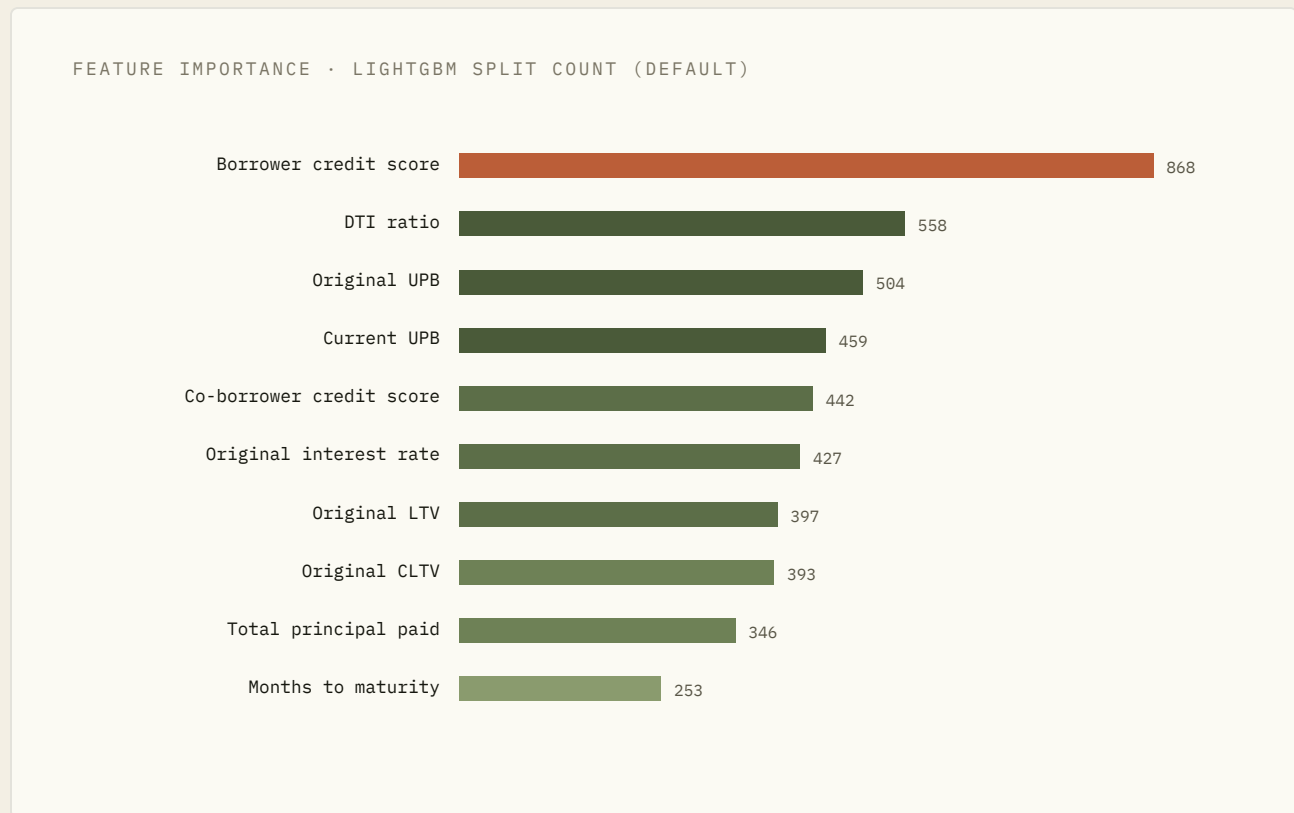


FIG 07. Top ten features by LightGBM's default split-count importance: how often each feature is used in a tree split. This metric favours high-cardinality numerics (credit score, DTI, balances) and understates the binary delinquency-status flags, which the EDA shows dominate near-term risk; a gain-based or SHAP ranking would lift those flags higher.

Conclusions & Recommendations

A recall-first delinquency model that turns 12.5 million monthly loan records into a short, workable watchlist, **catching seven in ten future defaults, six months ahead**, while asking a servicing team to review barely more than one percent of the book. The signals it leans on are the familiar credit fundamentals, which makes the model both effective and explainable to the risk teams who would rely on it.

WHERE IT GOES NEXT

01 Calibrate probabilities so the score can feed reserve sizing, not just ranking.

02 Add macro context (rates, regional unemployment) to harden against portfolio shift.

03 Tie the threshold to capacity, setting the cutoff from how many loans outreach can actually work.

04 Monitor drift & retrain cadence as newer origination vintages season.

"71% of future defaults caught, 1.4% of the book flagged, six months before the loans go bad."

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